Decentralized curation

Putting self-interest to work for the public good.

Decentralized Curation research by Ferit Tunçer.
Institute of Open Science Practices (IOSP) 2026, Leiden

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Who judges research?
What do we reward?

Read → Cite → Reuse → Fund

Every step depends on judgment.

One framework.
Here, public-goods funding.

Set the policy Pool creator

Nominate
Claims + bond

→

Judge relevance
Drawn jury

→

Fund
Relative weights

Accuracy: challenge claims anytime active

Relevance: appeal + re-score

Start with a public policy.

How much does this nominee's completed and maintained work underpin how research is done, shared and kept, now?

0.00	0.25	0.50	0.75	1.00
not infrastructure	matters to one field	relied on in several fields	widely relied on	critical across research

Evidence + Criteria + Fixed terms

This pool's authored policy, not a universal rubric · Policy reference

A claim you can check.

NUMPY 2.0.0

> “Released on 16 June 2024.”

Release record ↗

Delivery + Downstream use + Migration costs

Real release; illustrative nomination and scores · Release notes · Migration guide

One jury.
Different judgments.

6 seats · 3 curators

Curator A 3 tickets — 0.700



Curator B 2 tickets — 0.400



Curator C 1 tickets — 0.700



STAKE-WEIGHTED MEAN

0.600

(3 × 0.70 + 2 × 0.40 + 1 × 0.70) ÷ 6 = 0.600

Commit → reveal

Illustrative reports; one curator may hold several seats · Grading is separate

Disagree?
Fund another hearing.

INITIAL JURY

6

seats · score **0.600**

→

CONTESTED APPEAL

12

seats · score **0.700**

8 at 0.800; 4 at 0.500

Public backers. Larger jury. A revisable score.

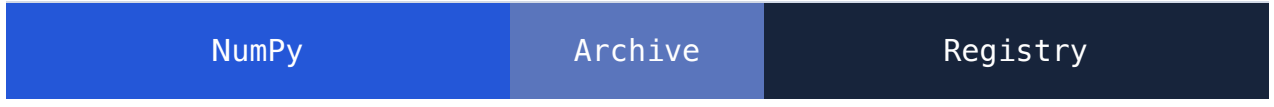
Intended crowdfunding design; an unopposed funded score wins by forfeiture · Details

A weight.
Not a percentage.

`effective weight = min(requested weight, adopted score)`

ILLUSTRATIVE POOL · 100 UNITS/DAY

Nominee	Request	Score	Weight	Units/day
NumPy	0.800	0.600	0.600	**40.00**
Archive	0.700	0.300	0.300	**20.00**
Registry	0.600	0.900	0.600	**40.00**



NUMPY'S WEIGHT

0.600

100 × 0.600 ÷ 1.500 = 40.00 units/day

NEW NOMINATION, AFTER ADMISSION HOLDBACK

After applicable holdback; relative allocation, not a measured ratio of public benefit.

Facts can be challenged.

COUNTERFACTUAL CLAIM

> “NumPy 1.x C extensions work unchanged in 2.0.”

These extensions require rebuilding ↗

Challenge → External arbitrator →

Debunked?

Bond lost; future funding stops

Illustration, not a verdict about NumPy. Previously earned funding remains owed.

New judgment.
New funding shares.

INITIAL SCORE

0.600

LATER SCORE

0.300

NUMPY / DAY

25.00

Lower weight takes effect immediately. Earlier accrual stays earned; only future accrual uses the new shares.

Illustrative re-score; return to the allocation.

Rules can be enforced.
Judgment must earn trust.

01

A DEFENSIBLE POLICY

02

USABLE EVIDENCE

03

INDEPENDENT SCRUTINY

Agreement alone is not enough.

Now you are the jury.



`https://decentralized-curation.0xferit.workers.dev/r/leiden`

Name → seat → score

No wallet. Workshop points only.

Judge the work.
Use the same question.

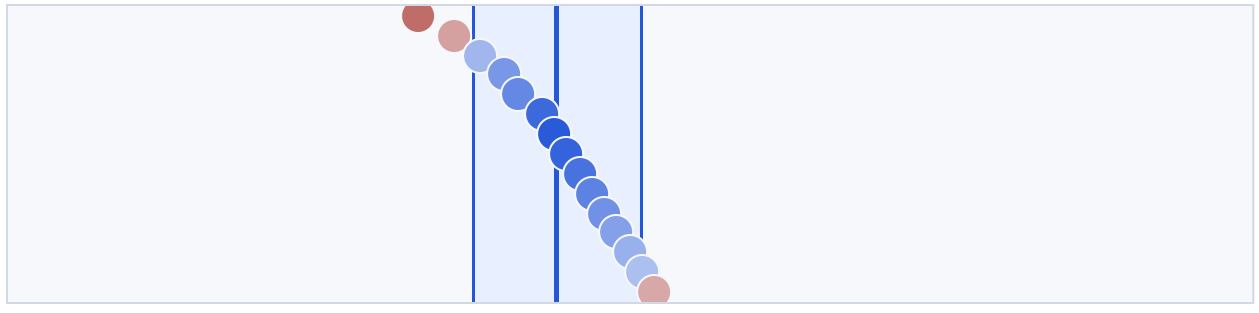
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how many rely · role in research · other options

Brief descriptions + prior knowledge; recusal is welcome. Full policy and exclusions

Your reports move the reference.



centre	spread	band	inside	forfeiting
0.44	0.055	±0.068	12	3

Blue: inside. Red: outside.

CENTRE

Weighted mean

BAND

Spread of reports

POINTS

Closeness inside the band

Workshop simplification; the implementation grades differently. Comparison

6 nominees.
Your judgment.

01
Inkscape

02
Linux

03
PDF

04
LaTeX

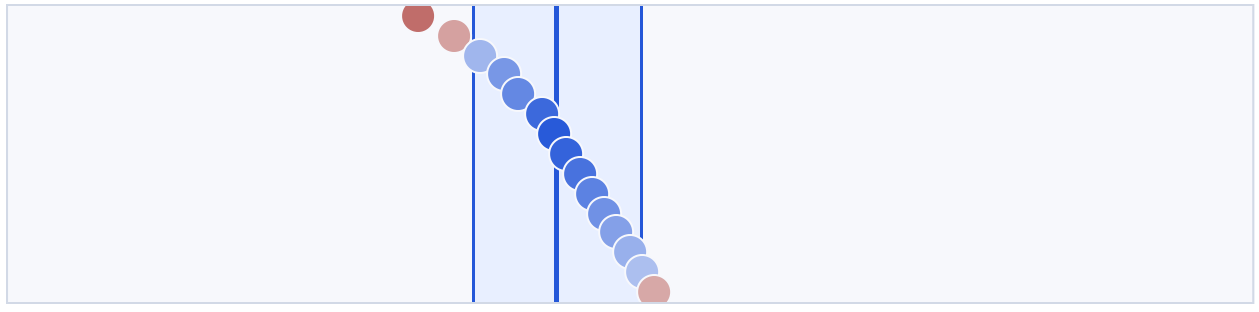
05
arXiv

06
NumPy

Read → Score alone → Seal → Open together

Keep your salt words private. Explain the evidence behind your score.

Agreement ≠ accuracy.



What did we reward?

What shaped the reference?

What would change your mind?

One room and an assigned bloc do not establish a safe collusion threshold.

Every defense has a boundary.

Attack	Response	Still exposed
Shift the criteria	Public policy	Bad policy; shared bias
Copy a score	Commit → reveal	Early reveals; coordination
Capture the jury	Stake-backed draw	Concentrated stake
Keep a bad ruling	Publicly funded appeal	Cost; shared bias
Prove a sealed score to a buyer	Workshop only: penalty + bounty	Other verification routes remain; no corresponding contract penalty found

Crowdfunded appeals: intended design. Other boundaries: inspected implementation. Details

Start smaller.
Learn before scaling.

HERE

WORKSHOP

Experience the game

→

PLANNED PRECURSOR

ARROW

Invited experts
Real funding docket

→

DESTINATION

FULL SYSTEM

Open participation
Adversarial conditions

Arrow assumes honesty, competence and non-collusion. No experiment results are claimed.

What would change our minds?

01

UNDERSTAND THE POLICY

02

RESPOND TO EVIDENCE

03

IMPROVE JUDGMENT

04

DELIVER PUBLIC BENEFIT

Earning more is not the success criterion.

Make scrutiny worthwhile.

Better shared judgment.
Better consequences.

doi.org/10.5281/zenodo.23313625 https://decentralized-curation.0xferit.workers.dev/join/

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Keep three scopes separate.

FULL DESIGN

Open, stake-backed juries
Accuracy + appeals
Continuing funding

ARROW PROPOSAL

Invited experts
Trusted nominations
Bounded experiment

THIS WORKSHOP

People in this room
Points
Teaching exercise

Intended design, implementation evidence and experiments are different claims.

Funding score ≠ curator grading.

FUNDING

ADOPTED SCORE

Jury ruling or forfeiture

INSPECTED IMPLEMENTATION

MEDIAN + DEVIATION

Final jury grades all rounds

WORKSHOP

MEAN + SPREAD

Immediate points

Deviation: median absolute deviation. Workshop uses standard deviation. Full comparison

Different tools.
A shared assessment problem.

DISCOVERY

MATADISCO

EXPLICIT CRITERIA

ASSESSMENT WORKSHOP

PUBLIC EXPERT REVIEW

THE UNJOURNAL

METRICS + HUMAN REVIEW

OPEN SOURCE ENDOWMENT

Potential complements; no comparative superiority or partnership is claimed.

Read the mechanism with its version.

1. doi.org/10.5281/zenodo.23313625

   Published preprint · version 1.2.0

2. Inspected implementation · ca8c856

   Access-controlled source snapshot; no deployment or audit claim

3. Arrow experiment proposal

   Plan, not results

4. Current mechanism and evidence summary

   Includes unpublished design clarifications

Detailed speaking notes: parallel speech document.